Software Requirements Specification for Flexin: subtitle describing software

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

1 Purpose of the Project

1.1 User Business

Insert your content here.

1.2 Goals of the Project

Insert your content here.

2 Stakeholders

This section will look at all individuals and groups that will affected by the success of this project. We will look further and study the people affected through the following sections.

2.1 Client

The client with whom the team is building the app is **Chris Schankula**. Chris was responsible for providing the initial idea and will work throughout each stage of the project to deliver the final product. *Insert your content here.*

2.2 Customer

- **Primary**: Young adults aged 18-30 who are seeking to build financial literacy and sustainable fitness habits.
- **Secondary:** Adults older than the intended age range who are looking to learn new skills, reinforce or start new healthy habits.

Insert your content here.

2.3 Other Stakeholders

Insert your content here.

2.4 Hands-On Users of the Project

- **Mobile (Android) Smartphone Users:** users will be able to interact with the app and utilize the different features to view and track financial and fitness goals. For the purpose of Capstone our final deployed app will be for Android users only.

Insert your content here.

2.5 Personas

2.5.1 Brandon - The Novice Saver and Avid Runner

Background

Brandon is a university student in his final year of Environmental Sciences at the University of Toronto, preparing for graduation in the spring. Brandon has recently begun learning more about his own finances, but still has a limited understanding of investing and saving.

Goals and Motivations

Brandon enjoys recreational running as a way to stay active, relieve stress, and maintain a healthy lifestyle. He wants to improve running performance, build consistency, and take this hobby to the next level. At the same time, Brandon wants to develop his financial literacy and gain confidence in investment decisions before investing real savings.

Behaviours and Habits

Brandon balances school responsibilities with running and personal development. He currently uses his phone as his main device to track his running activities such as goals and progress. Brandon prefers simple explanations, interactive learning, and measurable progress rather than complicated financial terminology.

Needs and Expectations

- A risk-free stock trading simulator that uses virtual money.

- Beginner-friendly lessons explaining investment opportunities, market trends, risk, and investment strategies.
- Progress tracking and feedback to help build investing knowledge and confidence.
- Running features to record workouts, monitor distance and pace, and track improvements over time.
- Achievable goals, milestones, and challenges that encourage consistent participation in both activities.

Frustrations

Brandon finds financial platforms intimidating because of unfamiliar terminology and the potential to lose money. Many investing resources assume prior knowledge, while running apps may focus primarily on recording workouts rather than helping beginners improve. Brandon wants a straightforward experience that makes learning and progress rewarding.

2.5.2 Connor – The Young Expert

Background

Connor is a third-year Commerce student specializing in Finance, returning to university after completing a year-long co-op placement at IG Wealth Management. Coming from a long line of financial professionals, Connor has developed a strong interest in financial management and aspires to build a successful career in the industry. As a top-performing student and hard-working individual, Connor is confident in his financial knowledge and enjoys applying what he has learned to real-world situations.

Goals and Motivations

Connor wants to excel academically and build on the professional experience gained during his co-op placement to advance his career in financial management. He enjoys demonstrating his financial knowledge and sharing his opinions on investment opportunities and market trends. Although Connor is an avid sports fan, he recognizes that balancing academics, career ambitions, and personal interests leaves him with little time to exercise. He wants

to find ways to incorporate physical activity into his busy schedule without compromising his professional goals.

Behaviours and Habits

Connor maintains a demanding schedule focused on academics, career development, and expanding his financial knowledge. He regularly follows financial news, monitors market trends, and enjoys discussing investment strategies with others. Connor is competitive and motivated by measurable results, academic achievement, and opportunities to demonstrate his expertise. While he enjoys watching sports, exercise is not currently a consistent part of his routine due to his time commitments.

Needs and Expectations

- Advanced financial tools and simulated trading features that allow him to apply his existing investment knowledge.
- Access to market trends, investment opportunities, and performance statistics to support informed decision-making.
- Competitive features such as rankings, challenges, and performance comparisons to showcase his financial expertise.
- Efficient running and exercise tracking that fits into his busy academic and professional schedule.
- Personalized fitness goals, progress tracking, and achievable challenges that encourage consistent physical activity.
- An engaging platform that combines financial competition with physical activity, allowing him to pursue both interests in one place.

Frustrations

Connor finds it difficult to balance his academic workload, career ambitions, and personal interests while maintaining an active lifestyle. Beginner-focused financial platforms may not provide enough depth or challenge to keep him engaged, while traditional fitness applications may not offer the competitive

elements that motivate him. Connor wants an efficient, challenging experience that allows him to demonstrate his financial knowledge while encouraging him to make time for exercise.

2.5.3 Daniella — The All-Arounder

Background

Daniella is a recent Civil Engineering graduate from the University of Guelph who gained valuable work experience and earned a substantial amount of money through multiple co-op placements throughout his degree. Through personal curiosity and working with her bank to understand her finances, Daniella developed a strong awareness of saving and managing money. Following graduation, she dedicated significant time to building her own investment portfolio and developing her financial knowledge. Although she has gained practical experience managing her savings, Daniella wants to become more confident in making independent investment decisions.

Goals and Motivations

Daniella enjoys staying active through a variety of organized sports which he participates in throughout the year. She worries that the time commitment required by multiple sports leagues may eventually become unsustainable, and makes the decision to cut back on some of the organized leagues that she participates in. She is interested in beginning a more traditional circuit-style workout as these tend to be more convenient. Financially, she wants to continue growing her investment portfolio while gaining more experience evaluating opportunities and making important investment decisions independently.

Behaviours and Habits

Daniella balances her professional responsibilities and personal interests while maintaining an active social life through organized sports. She prefers physical activities that are engaging, competitive, and social. Financially, Daniella takes an active role in monitoring his savings and managing his investment portfolio. She is curious, self-motivated, and willing to learn independently, but recognizes that managing a portfolio with real money offers limited op-

portunities to experiment with unfamiliar strategies without risking her savings.

Needs and Expectations

- A realistic stock trading simulator that allows her to practice making independent investment decisions without risking real money.
- Investment tools that help evaluate market trends, compare opportunities, and understand the potential risks and outcomes of different strategies.
- Performance tracking and feedback to evaluate investment decisions and identify areas for improvement.
- Convenient fitness tracking and personalized activity goals that fit her professional responsibilities and sports league commitments.
- Flexible exercise recommendations that provide alternatives to traditional gym workouts and help maintain fitness during busy periods.
- An integrated platform that encourages both financial development and physical activity without requiring excessive time or effort.

Frustrations

Daniella finds it challenging to balance his professional life, multiple sports leagues, and financial goals without over-committing her time. Although she enjoys sports, she recognizes that relying entirely on organized leagues may make it difficult to maintain a consistent fitness routine year-round. Similarly, while she has successfully developed her own investment portfolio, she wants more opportunities to test strategies and learn from her decisions without putting his savings at risk. Daniella wants a flexible and practical solution that supports her financial growth and physical well-being while fitting naturally into his lifestyle.

Insert your content here.

2.6 Priorities Assigned to Users

The project's main priority will be to focus on the beginner - novice users of the application. The goal will be to teach the basis of health/fitness and financial literacy. *Insert your content here.*

2.7 User Participation

User participation will be incorporated into the development process. User feedback will help to make design decisions and provide a better scope of features.

- User interviews planned for Term 1 (CD-Fall document) to help elicit requirements and plan for the development.
- User testing on demo builds planned for Term 2 (Weeks 17 and 21) to receive feedback on decisions and provide an outside source of feedback.

Insert your content here.

2.8 Maintenance Users and Service Technicians

Insert your content here.

3 Mandated Constraints

3.1 Solution Constraints

Insert your content here.

3.2 Implementation Environment of the Current System

Insert your content here.

3.3 Partner or Collaborative Applications

Insert your content here.

3.4 Off-the-Shelf Software

Insert your content here.

3.5 Anticipated Workplace Environment

Insert your content here.

3.6 Schedule Constraints

Insert your content here.

3.7 Budget Constraints

Insert your content here.

3.8 Enterprise Constraints

Insert your content here.

4 Naming Conventions and Terminology

4.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

Insert your content here.

5 Relevant Facts And Assumptions

5.1 Relevant Facts

Insert your content here.

5.2 Business Rules

Insert your content here.

5.3 Assumptions

Insert your content here.

6 The Scope of the Work

6.1 The Current Situation

The modern software landscape treats physical health management and financial education as separate domains, despite both relying on consistent daily actions and the long-term benefits of compounding. Existing fitness applications, such as Strava, Garmin Connect, and Google Fit, track physical activity through metrics such as steps, heart rate, and GPS routes, while using badges, streaks, and milestones to encourage engagement. However, the long-term effectiveness of these incentives varies, and meaningful health benefits often require sustained participation over time **??**.

Similarly, financial simulators provide users with risk-free investment experiences but may rely on virtual capital that is disconnected from real-world effort and financial constraints. These environments can allow users to experiment with investing without experiencing the consequences of actual financial decisions. Furthermore, their educational value depends on users' motivation to learn, rather than providing a direct incentive to develop stronger personal financial management habits.

Financial literacy remains a significant challenge in Canada. Boisclair et al. found that only 42% of Canadian respondents correctly answered three foundational questions concerning compound interest, inflation, and risk diversification **?**. The same study found that respondents who answered all three questions correctly were 10 percentage points more likely to have retirement savings, demonstrating an association between financial knowledge and retirement preparedness **?**.

Despite their shared emphasis on consistency and compounding, physical activity tracking and financial education remain largely disconnected. Fitness data is primarily managed through health-focused platforms, while investment simulators do not typically connect simulated capital to users' real-world effort or habit formation. Consequently, users lack an integrated way to explore how small, repeated actions can contribute to long-term outcomes across both domains.

This gap presents an opportunity for an integrated platform that connects physical activity with simulated investing. By linking consistent activity to incremental virtual investment opportunities, such a platform could introduce more tangible short-term incentives, demonstrate the effects of compounding, and teach financial concepts through familiar daily habits. This approach could encourage sustained participation while helping users develop the financial literacy needed for informed long-term decision-making.

6.2 The Context of the Work

Insert your content here.

6.3 Work Partitioning

Insert your content here.

6.4 Specifying a BUC

This section specifies each Flexin Business Use Case (BUC), including its goal, actors, trigger, preconditions, main flow, alternative flows, and post-conditions.

BUC 1: User Onboarding

Goal: Create a user account and initialize their Flexin profile.

Actors: End User.

Trigger: The user opens Flexin and selects the registration option.

Preconditions: The application is available and the user does not already have an active account.

Main Flow:

1. The user submits the required registration details.
2. Flexin validates the details and creates the account.
3. The user provides any requested baseline goals or preferences.
4. Flexin initializes the user's profile and virtual wallet.

Alternative Flows: If required details are missing or invalid, Flexin displays an error and requests corrections. If an account already exists, the user is prompted to sign in.

Postconditions: The user account and initial virtual wallet are created successfully.

BUC 2: Activity Ingestion

Goal: Retrieve and validate fitness activity data from supported integrations.

Actors: End User, Android Health Connect, Garmin Integration.

Trigger: New activity data becomes available or the user initiates an activity data synchronization.

Preconditions: The user is registered and has granted the required permissions to an available supported integration.

Main Flow:

1. Flexin retrieves available activity data from Health Connect or Garmin.
2. Flexin validates the data and checks for duplicate activities.
3. Valid activities are stored in the user's activity history.
4. The application updates the user's activity summary.

Alternative Flows: If permissions are missing, the user is prompted to grant access. If an integration is unavailable or data is invalid, Flexin reports the issue without recording the invalid activity.

Postconditions: Valid activity data is stored without duplicate records.

BUC 3: Currency Allocation

Goal: Reward eligible physical activity with virtual currency.

Actors: End User, Flexin Reward System.

Trigger: A validated activity satisfies a configured reward threshold.

Preconditions: The user has an active account and the activity has been validated.

Main Flow:

1. Flexin evaluates the activity against the reward rules.

2. The system calculates the eligible virtual currency reward.

3. Flexin credits the user's virtual wallet and records the transaction.

4. The user can view the updated balance and reward details.

Alternative Flows: If the activity does not meet the reward threshold, no currency is awarded. If the activity has already been rewarded, Flexin prevents duplicate crediting.

Postconditions: The wallet balance and transaction ledger reflect any eligible reward.

BUC 4: Asset Purchase

Goal: Allow the user to purchase fictional financial assets using virtual currency.

Actors: End User, Flexin Market Engine.

Trigger: The user submits a simulated buy order.

Preconditions: The user is registered, the selected asset exists, and a valid simulated price is available.

Main Flow:

1. The user selects an asset and enters the purchase quantity.

2. Flexin calculates the simulated purchase cost.

3. The system verifies that the user has sufficient virtual currency.

4. Flexin executes the simulated order and updates the wallet and holdings.

5. The completed transaction is recorded and displayed to the user.

Alternative Flows: If the balance is insufficient, the order is rejected. If the asset or price is unavailable, the user is informed and the transaction is not executed.

Postconditions: For a successful purchase, the virtual balance is reduced and the user's asset holdings increase accordingly.

BUC 5: Asset Sale

Goal: Allow the user to sell fictional assets and receive virtual currency.

Actors: End User, Flexin Market Engine.

Trigger: The user submits a simulated sell order.

Preconditions: The user holds a sufficient quantity of the selected asset and a valid simulated price is available.

Main Flow:

1. The user selects an asset and enters the quantity to sell.
2. Flexin calculates the simulated sale proceeds.
3. The system verifies that the user holds sufficient units.
4. Flexin executes the simulated order and updates the holdings and wallet.
5. The completed transaction is recorded and displayed to the user.

Alternative Flows: If the user holds insufficient units or the asset price is unavailable, the order is rejected without changing the portfolio.

Postconditions: For a successful sale, the user's holdings decrease and the corresponding virtual currency is credited.

BUC 6: Market Simulation

Goal: Generate and update fictional asset prices and market events.

Actors: Flexin Market Engine, Flexin Scheduler.

Trigger: A scheduled market simulation cycle begins.

Preconditions: The market engine is operational and fictional assets have been initialized.

Main Flow:

1. The scheduler initiates a simulation cycle.
2. The engine generates price changes and fictional market events according to the system's simulation rules.
3. Flexin updates asset valuations and relevant market information.

4. Updated prices and events become available to portfolio and trading functions.

Alternative Flows: If a simulation cycle fails, Flexin records the failure and retains the last valid market state until an update succeeds.

Postconditions: The latest valid fictional asset prices and market events are available within Flexin. The engine does not rely on real-world financial market data.

BUC 7: Portfolio Analytics

Goal: Present the user with portfolio performance and activity-related progress.

Actors: End User.

Trigger: The user opens the portfolio analytics view.

Preconditions: The user is registered and portfolio or activity data is available.

Main Flow:

1. Flexin retrieves the user's virtual balance, holdings, and transaction history.
2. The system calculates portfolio value and performance metrics using simulated asset prices.
3. Flexin retrieves relevant activity and reward information.
4. The application displays the resulting portfolio and activity summaries.

Alternative Flows: If insufficient history exists, Flexin displays available information and indicates that more data is needed for certain metrics.

Postconditions: The user can view their latest available portfolio and activity analytics.

BUC 8: Financial Education

Goal: Provide educational guidance relevant to the user's simulated investing activity.

Actors: End User, Flexin Educational Rules Engine.

Trigger: The user requests financial guidance or opens an educational feature.

Preconditions: The user can access the application.
Main Flow:

1. Flexin identifies the context of the user's request.
2. The educational rules engine selects relevant educational content.
3. Flexin presents explanations or guidance to the user.
4. The user reviews the content and may return to the application.

Alternative Flows: If no specific guidance matches the request, Flexin presents general educational content.
Postconditions: Relevant educational content is displayed. Guidance is educational and does not execute trades on the user's behalf.

BUC 9: Community Leaderboard

Goal: Allow users to view community rankings based on eligible activity or simulated portfolio metrics.
Actors: End User.
Trigger: The user opens the community leaderboard.
Preconditions: The user has access to Flexin and has enabled any participation required for their own information to appear in rankings.
Main Flow:

1. Flexin retrieves eligible ranking data from participating users.
2. The system calculates or retrieves rankings using the configured metrics.
3. The application displays the leaderboard and its ranking criteria.

Alternative Flows: If the user has opted out or no ranking data is available, Flexin displays the available leaderboard information without including the user's opted-out data.
Postconditions: The user can view the available community rankings in accordance with participation and privacy settings.

BUC 10: Notifications

Goal: Notify users about relevant activity rewards, reminders, and simulated market events.

Actors: End User, Flexin Notification Service, Android Device and OS.

Trigger: A configured reminder, eligible reward, or relevant simulated market event occurs.

Preconditions: The relevant event or schedule exists, and the user has enabled the applicable notifications and granted any required device permissions.

Main Flow:

1. Flexin identifies the event and determines whether a notification should be sent.
2. The system generates a notification containing relevant information.
3. Flexin sends the notification through the supported device mechanism.
4. The user receives the notification if device settings and permissions permit delivery.

Alternative Flows: If notifications are disabled or delivery fails, Flexin does not guarantee delivery and may retain the event for in-app display.

Postconditions: The notification is submitted for delivery or the delivery failure is handled according to the system's notification rules.

7 Business Data Model and Data Dictionary

7.1 Business Data Model

Insert your content here.

7.2 Data Dictionary

Insert your content here.

8 The Scope of the Product

8.1 Product Boundary

Insert your content here.

8.2 Product Use Case Table

Insert your content here.

8.3 Individual Product Use Cases (PUC's)

Insert your content here.

9 Functional Requirements

9.1 Functional Requirements

Insert your content here.

10 Look and Feel Requirements

10.1 Appearance Requirements

Insert your content here.

10.2 Style Requirements

Insert your content here.

11 Usability and Humanity Requirements

11.1 Ease of Use Requirements

Insert your content here.

11.2 Personalization and Internationalization Requirements

ID	Requirement	Specification
UHR-PER-01	Personalized Fitness Goals	**Description:** The system shall allow users to configure fitness goals based on their personal fitness level and preferences. **Rationale:** Users will possess different fitness capabilities and activity levels. Personalized goals encourage engagement while remaining achievable. **Fit Criterion:** Users shall be able to modify daily and weekly fitness goals that will take effect the next time period.
UHR-PER-02	Customizable Notifications	**Description:** The system shall allow users to configure which notifications they receive and how frequently they are delivered. **Rationale:** Users have varying preferences regarding reminders and activity notifications. **Fit Criterion:** Users shall be able to independently enable or disable workout reminders, portfolio updates, and achievement notifications.
UHR-PER-03	Customizable Privacy Settings	**Description:** The system shall allow users to control the visibility of their fitness achievements, portfolio statistics, and leaderboard participation. **Rationale:** Users have different privacy expectations and comfort levels regarding public sharing of personal information. **Fit Criterion:** Users shall be able to independently configure visibility settings for all community-facing features that will take immediate effect.

ID	Requirement	Specification
UHR-PER-04	Customizable Competition Groups	**Description:** The system shall allow users to customize the set of users with whom they share and compare fitness and portfolio progress. **Rationale:** Users may have different comfort levels regarding competition and sharing personal achievements. Allowing users to select their competition group creates a more personalized and engaging experience where friends can motivate each other. **Fit Criterion:** Users shall be able to choose between private, friends-only, and public, community-wide, leaderboard participation.
UHR-INT-01	Canadian Time Zone Support	**Description:** The system shall automatically detect and utilize the user's local Canadian time zone when calculating daily goals, rewards, and streaks. **Rationale:** Users may access the application from any Canadian province or territory. **Fit Criterion:** Daily statistics and streak calculations shall correspond to the user's local device time zone.

11.3 Learning Requirements

ID	Requirement	Specification
UHR-LRN-01	Minimal User Training	**Description:** The system shall enable first-time users to successfully use the application's core functionality without requiring formal training. **Rationale:** The application targets users with varying expertises and should be intuitive for a broad audience. **Fit Criterion:** At least 80% of first-time users shall be able to synchronize fitness data, earn virtual currency, and purchase an investment asset without external assistance.
UHR-LRN-02	Learning Curve of Financial Concepts	**Description:** The system shall present investment concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on financial investing. This means that all users, regardless of their prior financial experience, should be capable of learning and understanding the financial concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time investors shall successfully complete their first virtual investment and view their portfolio's growth after one week without assistance.

ID	Requirement	Specification
UHR-LRN-03	Learning Curve of Fitness Based Concepts	**Description:** The system shall present fitness concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on fitness and health related activities. This means that all users, regardless of their prior fitness experience, should be capable of learning and understanding the fitness concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time participants in a new fitness activity shall successfully accomplish their set goal and view their health metrics after one week without assistance.
UHR-LRN-04	Integrated Educational Assistance	**Description:** The system shall provide contextual learning resources within the application. **Rationale:** In the event that users encounter unfamiliar fitness or financial terminology while using the platform, the system should offer immediate assistance to enhance their understanding so that users can quickly learn and pick up new concepts. **Fit Criterion:** Users shall be able to access explanatory information for all core financial and fitness concepts directly from the screen where the concept appears.

ID	Requirement	Specification
UHR-LRN-05	User Help Documentation	**Description:** The system shall provide readily accessible help documentation describing the application's functionality should any users require it. **Rationale:** Users should be able to resolve common questions without external support. **Fit Criterion:** A help section shall be accessible from the application's settings menu and include explanations of all major features. Support teams should receive zero inquiries about basic system functionality.

11.4 Understandability and Politeness Requirements

Insert your content here.

11.5 Accessibility Requirements

Insert your content here.

12 Performance Requirements

12.1 Speed and Latency Requirements

Insert your content here.

12.2 Safety-Critical Requirements

Insert your content here.

12.3 Precision or Accuracy Requirements

Insert your content here.

12.4 Robustness or Fault-Tolerance Requirements

Insert your content here.

12.5 Capacity Requirements

Insert your content here.

12.6 Scalability or Extensibility Requirements

Insert your content here.

12.7 Longevity Requirements

Insert your content here.

13 Operational and Environmental Requirements

13.1 Expected Physical Environment

The application will be operable through mobile smartphones and smart tablets with cellular/Wi-Fi connectivity. Inputs will be completed through health tracking services that will be provided by the smart device that is running the software, or optionally received through other technologies used for health tracking.

13.2 Wider Environment Requirements

Insert your content here.

13.3 Requirements for Interfacing with Adjacent Systems

Many outside systems will be used:

- HTTPS connection to external APIs
    1. External Technology Connection (e.g., Garmin API, Strava SDK)
    2. Software-specific API for health-related tracking (Android Health Connect)
- HTTPS connection to internal APIs
    1. Java SpringBoot API for storing user data and retrieving from the database

- Database Connection
    1. Privately hosted MySQL server

13.4 Productization Requirements

Insert your content here.

13.5 Release Requirements

Insert your content here.

14 Maintainability and Support Requirements

This section outlines the requirements for maintaining and supporting the software product throughout its lifecycle. It includes specifications for maintenance procedures, support services, and any necessary adaptability procedures to ensure the product remains functional and up-to-date.

14.1 Maintenance Requirements

ID	Requirement	Specification
MAS-MNT-01	Fitness Activity Addition Effort	**Description:** The system shall support the addition of new reward-eligible fitness activity types with minimal maintenance effort. **Rationale:** Future releases may introduce additional forms of exercise and adaptive fitness activities. **Fit Criterion:** A new activity type shall be implemented, deployed, and fully tested within 2 months.

ID	Requirement	Specification
MAS-MNT-02	Investment Asset Addition Effort	**Description:** The system shall support the addition of new simulated investment assets with minimal maintenance effort. **Rationale:** Future versions of the application may expand the types of simulated investments available to users. **Fit Criterion:** A new investment asset shall be implemented, deployed, and fully tested within 2 months.
MAS-MNT-03	Corrective Maintenance Effort	**Description:** The system shall efficiently handle and correct software defects. **Rationale:** Timely defect resolution improves reliability and user satisfaction. **Fit Criterion:** Critical production defects shall be fixed and deployed within 5 business days of of the defect being reported.
MAS-MNT-04	Third-Party Integration Maintenance	**Description:** The system shall support updates to external fitness platform integrations with minimal impact on overall system functionality. **Rationale:** External APIs may change independently of the application and require maintenance. **Fit Criterion:** Changes required to maintain compatibility with a supported fitness provider shall be deployed within 5 business days of identifying the incompatibility.

ID	Requirement	Specification
MAS-MNT-05	Traceability	**Description:** The system shall provide sufficient documentation and traceability to allow future developers to efficiently understand previous changes and maintenance activities. **Rationale:** Accurate and up-to-date documentation improves maintainability by allowing future developers to understand previous modifications, known issues, and the reasoning behind architectural and implementation decisions. **Fit Criterion:** For every production release, developers shall be able to identify the implemented changes, associated issues, and documented design decisions within 30 minutes of searching project records.

14.2 Supportability Requirements

Insert your content here.

14.3 Adaptability Requirements

ID	Requirement	Specification
MAS-ADPT-01	External Fitness Provider Adaptability	**Description:** The system shall support the addition of new fitness data providers without requiring major changes to user workflows. **Rationale:** The ecosystem of fitness tracking applications and devices will evolve over time and additional providers may become desirable. **Fit Criterion:** Users shall be able to successfully connect newly supported fitness providers using the same workflow used for existing providers.

ID	Requirement	Specification
MAS-ADPT-02	Mobile Platform Portability	**Description:** The system shall support future deployment on additional mobile platforms beyond the initially targeted platform - Android devices. **Rationale:** Initial deployment is limited to Android, but future releases may target additional mobile operating systems, such as iOS. **Fit Criterion:** User account information and application data shall remain portable across supported platforms.
MAS-ADPT-03	Future AI Service Adaptability	**Description:** The system shall support future integration of AI-assisted educational content. **Rationale:** With the rapid growth of AI technologies, future releases may provide personalized fitness and financial learning experiences using AI technologies. **Fit Criterion:** AI-powered educational services shall be deployable without requiring modifications to existing user accounts.

15 Security Requirements

15.1 Access Requirements

ID	Requirement	Specification
SEC-ACC-01	Authentication and Sessions	**Description:** Users shall authenticate before accessing personal fitness data, portfolios, or transactions. The React Native client shall authenticate with the Spring API using secure token-based sessions. **Rationale:** Authentication protects personal data and virtual assets from unauthorized access. **Fit Criterion:** All protected API endpoints shall reject missing or invalid access tokens with HTTP 401. Access tokens shall expire within 60 minutes, with refresh or re-authentication required.
SEC-ACC-02	Role-Based Authorization	**Description:** Role-based access control shall restrict users to their own records and limit administrative or evaluator access to authorized functions. **Rationale:** Access restrictions prevent unauthorized modifications and cross-account data exposure. **Fit Criterion:** JUnit security tests shall confirm that attempts by standard users to access or modify another user's records return HTTP 403 in all tested cases.
SEC-ACC-03	Fitness Integration Authorization	**Description:** Flexin shall request only necessary read permissions for supported integrations, Android Health Connect and Garmin. Integration credentials shall be stored using secure, Android Keystore-backed storage. **Rationale:** Secure credential handling protects external accounts and limits access to fitness data. **Fit Criterion:** Integration tokens shall use Android Keystore-backed secure storage. No credentials, tokens, or secrets shall be committed to the project repository.

15.2 Integrity Requirements

ID	Requirement	Specification
SEC-INT-01	Activity Validation	**Description:** The Spring backend shall validate activity metrics and detect duplicate records, overlapping timestamps, and implausible activity values before awarding virtual currency. **Rationale:** Validation prevents activity manipulation and unauthorized currency generation. **Fit Criterion:** Automated tests shall confirm that duplicate or flagged implausible activity records are marked unverified and receive zero virtual currency. Sustained cadence above 250 steps per minute shall be flagged for verification.
SEC-INT-02	Ledger Consistency	**Description:** All virtual currency operations, including rewards, trades, and dividends, shall use ACID-compliant MySQL transactions. **Rationale:** Transactional consistency prevents duplicate rewards, negative balances, and partial updates. **Fit Criterion:** Stress tests with 100 concurrent trade requests against one wallet shall verify that every operation commits consistently or rolls back completely, without partial state changes.

ID	Requirement	Specification
SEC-INT-03	Server-Side Valuation	**Description:** The Spring backend and custom market engine shall determine asset prices, portfolio valuations, and currency calculations. The mobile client shall not control authoritative values. **Rationale:** Server-side calculations prevent client manipulation of prices and balances. **Fit Criterion:** Trade requests attempting to specify unauthorized execution prices shall be rejected with HTTP 400.

15.3 Privacy Requirements

ID	Requirement	Specification
SEC-PRV-01	Health Data Protection	**Description:** Sensitive personal information and fitness telemetry shall be protected through encryption in transit and at rest. **Rationale:** Encryption protects confidential user information against unauthorized disclosure. **Fit Criterion:** Client-server communications shall use TLS 1.3. Sensitive biometric data stored in MySQL shall be encrypted at rest using AES-256.

ID	Requirement	Specification
SEC-PRV-02	Opt-In Community Sharing	**Description:** Leaderboard participation shall be opt-in. Users may share activity streaks and simulated portfolio returns using a chosen alias. **Rationale:** Users should control whether their activity and simulated financial performance are visible to others. **Fit Criterion:** Leaderboard sharing shall be disabled by default. Enabling it shall require explicit consent and disclose which pseudonymous information and statistics will be shared.
SEC-PRV-03	User Data Deletion	**Description:** Users shall be able to request account deletion through the application settings, removing associated personal records. **Rationale:** Users should retain control over their personal information and account history. **Fit Criterion:** Personal health metrics, credentials, and fitness history shall be purged from the active database within 24 hours of a deletion request. Only appropriately anonymized transaction aggregates may be retained for auditing.

15.4 Audit Requirements

ID	Requirement	Specification
SEC-AUD-01	Economic Event Logging	**Description:** Flexin shall maintain an append-only audit ledger for currency rewards, simulated trades, dividends, and market price updates. **Rationale:** Auditable records support troubleshooting, transaction verification, and evaluation of the simulated economy. **Fit Criterion:** Database permissions shall prevent UPDATE and DELETE operations on audit records. Every recorded transaction shall reference a valid source activity or market event, where applicable.
SEC-AUD-02	Security Event Logging	**Description:** The backend shall log failed authentication, token validation failures, duplicate activity submissions, anti-exploit flags, and runtime exceptions. **Rationale:** Security logs help identify suspicious activity and diagnose integration or system failures. **Fit Criterion:** Structured logs shall capture severity, client IP address where available, authenticated user ID where applicable, and exception details for runtime errors.

15.5 Immunity Requirements

ID	Requirement	Specification
SEC-IMM-01	Injection Defense	**Description:** The backend and database layer shall be protected against SQL injection, cross-site scripting (XSS), and command injection. **Rationale:** Injection vulnerabilities could compromise database integrity and expose user information. **Fit Criterion:** Database access shall use parameterized queries or prepared statements. Static analysis and security reviews shall identify zero unresolved high-severity injection vulnerabilities before release.
SEC-IMM-02	Rate Limiting	**Description:** The Spring API shall limit request rates for authentication, trade submission, and activity synchronization endpoints. **Rationale:** Rate limiting reduces brute-force attempts, automated abuse, and service overload. **Fit Criterion:** A configured limit, initially 60 requests per minute per authenticated user or client IP, shall be enforced. Excess requests shall receive HTTP 429. Automated tests shall verify enforcement.

ID	Requirement	Specification
SEC-IMM-03	Client Environment Validation	**Description:** The Android client shall detect supported indicators of rooted or otherwise untrusted environments and restrict sensitive functionality when tampering is detected outside developer builds. **Rationale:** Compromised environments can increase the risk of credential exposure and activity manipulation. **Fit Criterion:** In supported production configurations, detected root or tampering indicators shall trigger restrictions on community leaderboard access and virtual currency minting. Developer builds shall remain available for authorized testing.

16 Cultural Requirements

This section outlines the cultural requirements for the software system. Cultural requirements refer to the expectations and norms of the target user base, which may vary based on geographic location, language, social norms, and other cultural factors. These requirements ensure that the software is culturally sensitive and appropriate for its intended audience.

16.1 Cultural Requirements

ID	Requirement	Specification
CUL-01	Supports Canadian Languages	**Description:** The system shall present all user-facing functionality with the option to choose between English and French. **Rationale:** The application will initially be deployed strictly within Canada, where English and French are the country's official languages. **Fit Criterion:** All user-facing text shall be stored separately from application logic and be replaceable through a localization framework.
CUL-02	Inclusive Financial Education	**Description:** The system shall provide financial education content that does not assume prior investing knowledge. **Rationale:** Users possess varying levels of financial literacy. The system seeks to educate users of all levels of financial literacy - including those who may have limited exposure to investing concepts. **Fit Criterion:** All investment-related terminology presented to users shall include a plain-language explanation accessible from within the application. All financial concepts shall be explained in simple, non-technical language to start, until the system deems the user has gained sufficient experience within the application. During user testing, a survey will be conducted to gather feedback on the clarity and understandibility of the financial education content - at least 80% of users shall express great satisfaction with little to no problems understanding financial concepts.

ID	Requirement	Specification
CUL-03	Inclusive Fitness Participation	**Description:** The system shall accommodate users with varying fitness levels by allowing personalized activity participation. **Rationale:** Users may differ significantly in age, physical ability, health conditions, and exercise experience. The application should promote participation regardless of fitness background. **Fit Criterion:** Users shall be able to participate in their preferred fitness activities while still gaining subsequent rewards.
CUL-04	Culturally Neutral Educational Content	**Description:** The system shall present fitness and financial education content using culturally neutral language and examples (i.e. sample companies in the virtual market). **Rationale:** The Canadian population consists of individuals from diverse cultural, religious, and socioeconomic backgrounds. Educational content should be inclusive and broadly applicable. **Fit Criterion:** Educational content shall not require familiarity with a specific cultural background, religion, ethnicity, or region to be understood.

17 Compliance Requirements

17.1 Legal Requirements

Insert your content here.

17.2 Standards Compliance Requirements

ID	Requirement	Specification
COMP-STD-01	Android Platform Compliance	**Description:** The system shall comply with the Google Play Developer Program Policies. **Rationale:** The application is intended to be deployed on the Google Play Store as to be accessible on Android devices and as a result must conform to platform requirements. **Fit Criterion:** The application shall successfully pass all mandatory checks required for distribution through the Google Play Store.
COMP-STD-02	Accessibility Compliance	**Description:** The system shall comply with WCAG 2.1 Level AA accessibility guidelines for user-facing functionality. **Rationale:** The system should offer a simple and delightful user experience that should be accessible to users with varying abilities (i.e. user's with colour-blindness, hearing impairments, etc.). **Fit Criterion:** All major user workflows shall satisfy WCAG 2.1 Level AA success criteria as evaluated using automated accessibility tools and manual testing.
COMP-STD-03	TypeScript Coding Standards Compliance	**Description:** The frontend codebase shall comply with the Google TypeScript Style Guide. **Rationale:** Consistent coding standards improve maintainability, readability, and long-term project quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.

ID	Requirement	Specification
COMP-STD-04	Java Coding Standards Compliance	**Description:** The backend codebase shall comply with the Oracle Java Code Conventions. **Rationale:** Standardized coding practices improve maintainability, code review effectiveness, and software quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.
COMP-STD-05	Database Compliance	**Description:** The system shall store data in compliance with the MySQL Reference Manual and the SQL Standard (ISO/IEC 9075). **Rationale:** The project architecture relies upon MySQL as the primary relational database system. **Fit Criterion:** All schema definitions and queries will be tested to execute successfully on the supported MySQL version without compatibility warnings or errors.
COMP-STD-06	Software Testing Standard Compliance	**Description:** The system shall undergo thorough and complete testing in accordance with the IEEE 29119 Software Testing Standard. **Rationale:** Consistent testing practices reduce defects and improve software reliability. The system will comply with this standard to provide a unified, process-driven framework for software testing. **Fit Criterion:** All code merged into the main branch shall pass automated test execution in the CI/CD pipeline based on tests that were developed in accordance with the IEEE 29119 Software Testing Standard.

ID	Requirement	Specification
COMP-STD-07	Information Security Compliance	**Description:** The system shall be designed in accordance with the security principles outlined in ISO/IEC 27001. **Rationale:** The application stores personal, financial-learning, and fitness-related information that must be protected against unauthorized access and disclosure. **Fit Criterion:** Security controls for authentication, access control, data storage, and data transmission shall align with ISO/IEC 27001 recommendations.

18 Open Issues

Insert your content here.

19 Off-the-Shelf Solutions

19.1 Ready-Made Products

Insert your content here.

19.2 Reusable Components

Insert your content here.

19.3 Products That Can Be Copied

Insert your content here.

20 New Problems

20.1 Effects on the Current Environment

Insert your content here.

20.2 Effects on the Installed Systems

Insert your content here.

20.3 Potential User Problems

Insert your content here.

20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Insert your content here.

20.5 Follow-Up Problems

Insert your content here.

21 Tasks

21.1 Project Planning

Insert your content here.

21.2 Planning of the Development Phases

Insert your content here.

22 Migration to the New Product

22.1 Requirements for Migration to the New Product

Insert your content here.

22.2 Data That Has to be Modified or Translated for the New System

Insert your content here.

23 Costs

Insert your content here.

24 User Documentation and Training

24.1 User Documentation Requirements

Insert your content here.

24.2 Training Requirements

Table 1: Flexin Training Requirements

ID	Requirement	Specification
TRN-01	Zero-Training Design	**Description:** Flexin shall follow a walk-up-and-use design requiring no mandatory formal training or external assistance. **Rationale:** Intuitive navigation reduces onboarding effort and encourages continued use. **Fit Criterion:** At least 90% of first-time test users shall register, connect a fitness data source, and view their dashboard within 3 minutes without assistance.
TRN-02	Interactive Onboarding	**Description:** After account creation, Flexin shall offer an optional, dismissible four-step walkthrough covering fitness goals, activity rewards, simulated investing, and the compounding chart. **Rationale:** Guided interaction introduces the application's core features through practical examples. **Fit Criterion:** The walkthrough shall take under 90 seconds, provide a visible "Skip Tour" option on every step, and remain accessible through settings.

Continued on next page

Table 1: Flexin Training Requirements (Continued)

ID	Requirement	Specification
TRN-03	Progressive Feature Discovery	**Description:** Flexin shall progressively introduce advanced features, including portfolio rebalancing, volatility filters, and leaderboard sharing, as users gain experience. **Rationale:** Gradual feature discovery reduces cognitive overload for novice users. **Fit Criterion:** Advanced trading indicators and portfolio analytics shall remain collapsed or secondary until the user completes an initial trade or logs at least three days of physical activity.
TRN-04	Demonstrator Quick-Start Kit	**Description:** The team shall provide a demonstrator guide and automated demo-data seeding script for project evaluations and presentations. **Rationale:** Preloaded data enables evaluators to verify application features without completing real-time workouts. **Fit Criterion:** A preset configuration or debug toggle shall load 30 days of simulated activity and portfolio history within 10 seconds, enabling demonstrations of trading and compounding charts.

25 Waiting Room

Insert your content here.

26 Ideas for Solution

Insert your content here.

27 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

27.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

27.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

27.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

28 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?